

THE EFFECT OF DAMPING ON THE VIBRATION-STABILITY RELATION OF LINEAR SYSTEMS

D.E. Beskos

Department of Civil and Mineral Engineering, University of Minnesota, Minneapolis, Minnesota 55455, U.S.A.

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Introduction

The relation between flexural vibration and elastic stability for linear elastic systems without damping has been investigated in the past. Thus, Lurie [1], dealing with beams and plates of continuous mass distribution, showed that the load corresponding to zero frequency is the buckling load. Furthermore, he pointed out that there is a linear relationship between the square of the frequency of flexural vibration and the end load, if the mode of free vibration is identical to the buckling mode. The same results were found to be also true for linear discrete elastic systems of a finite number of degrees of freedom (e.g. in [2]).

In this paper the damping of the system is taken into account, and its influence on the previous results is studied. Two kinds of damped systems are considered: Beams and plates with continuous mass distribution and a linear elastic discrete system of a finite number of degrees of freedom.

Analysis

The equation of free damped lateral motion of a uniform beam under a constant axial compressive force P is given by [3]

$$EI \frac{\partial^4 y(x,t)}{\partial x^4} + P \frac{\partial^2 y(x,t)}{\partial x^2} + c \frac{\partial y(x,t)}{\partial t} + m \frac{\partial^2 y(x,t)}{\partial t^2} = 0, \quad (1)$$

where EI is the bending rigidity, c is the viscous damping coefficient, and m is the mass per unit of length of the beam. By assuming a solution of the form

$$y(x,t) = Y(x)e^{\bar{p}t}, \quad (2)$$

equation (1) reduces to the system of equations

$$EI \frac{\partial^4 Y}{\partial x^4} + P \frac{\partial^2 Y}{\partial x^2} - \bar{w}_{on}^2 mY = 0, \quad (3)$$

$$\bar{p}_n = -\beta \pm i\bar{w}_n, \quad \bar{w}_n^2 = \bar{w}_{on}^2 - \beta^2, \quad \beta = c/2m. \quad (4)$$